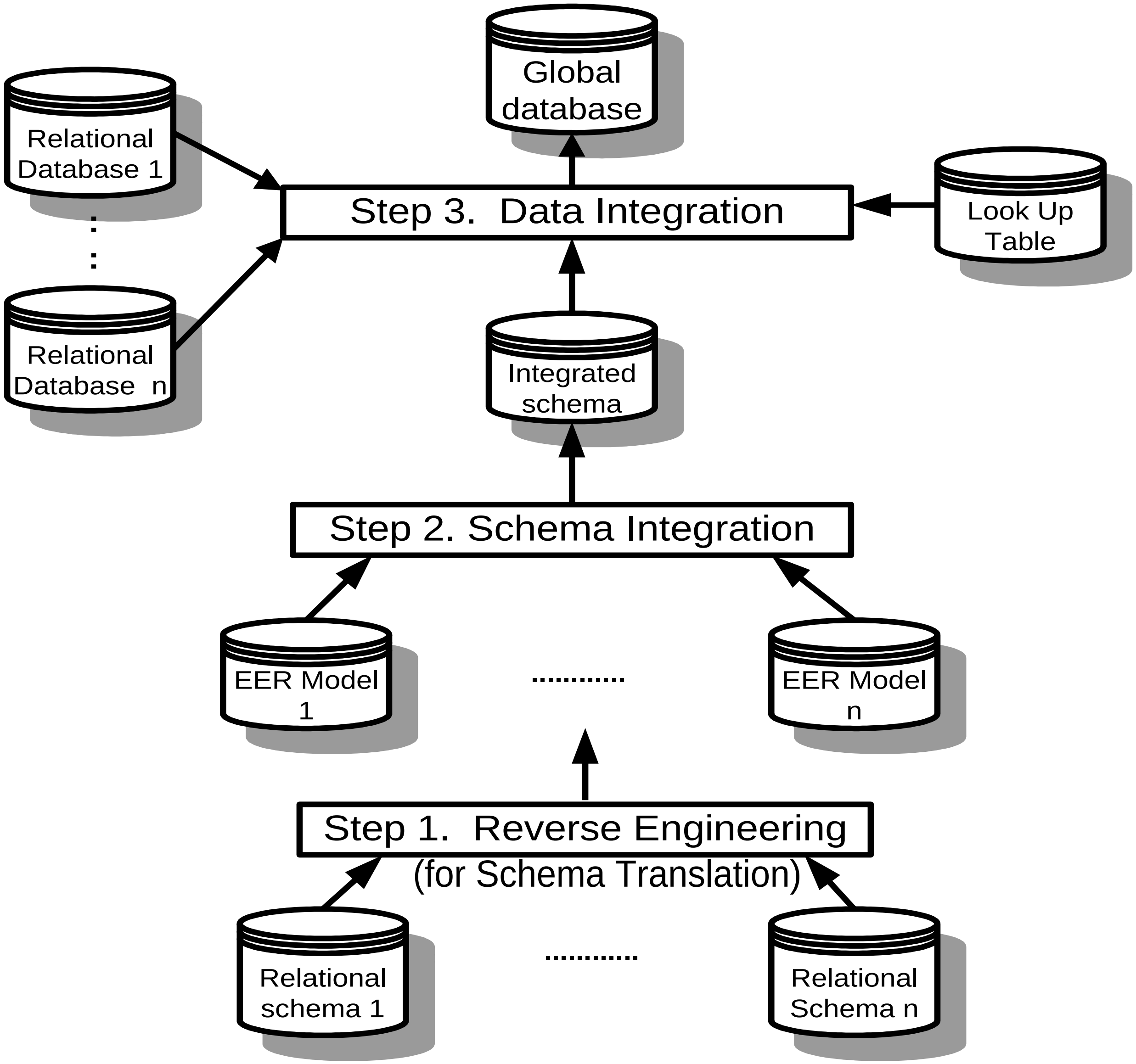
Architecture of multiple databases integration



Data Integration Concepts

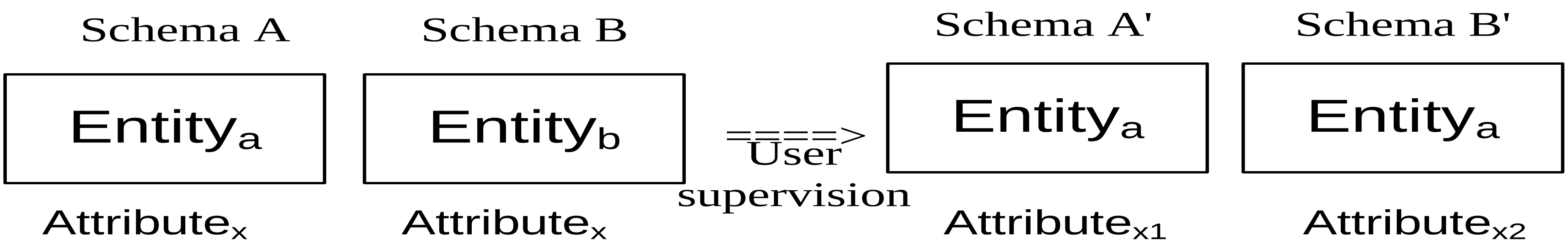
- Data semantics define the relationships between data for user’s data requirements.
- Data semantics are presented in database conceptual schema such as EER model and DTD Graph.
- Only relevant data can be integrated for an application.
- Data relevance depends on user’s data requirements for an application.
- Data consistency are the standard of data domain value and format. Inconsistent data must be transformed before data integration.
- User supervision means users input for user’s data requirements, and which are for database design and schema integration

Schema integration of EER models

```
Begin For each existing database do
  Begin If  its conceptual schema does not exist
        then reconstruct its EER model by reverse engineering;
       For each pair of existing databases' EER models schema A and schema B do
      begin resolve semantic conflicts between EER model A and EER model B;/step1/
            Merge classes between EER model A and EER model B;           /*step2*/
            Merge relationships between EER model A and EER model B;     /*step3*/
        end
    end
```

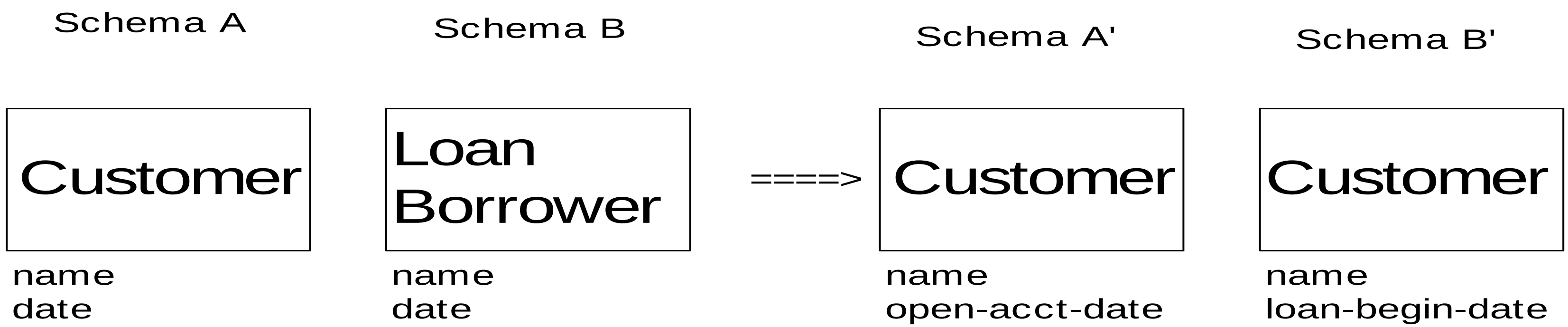
Step 1 Resolve conflicts among EER model

Resolve conflicts on synonyms and homonyms

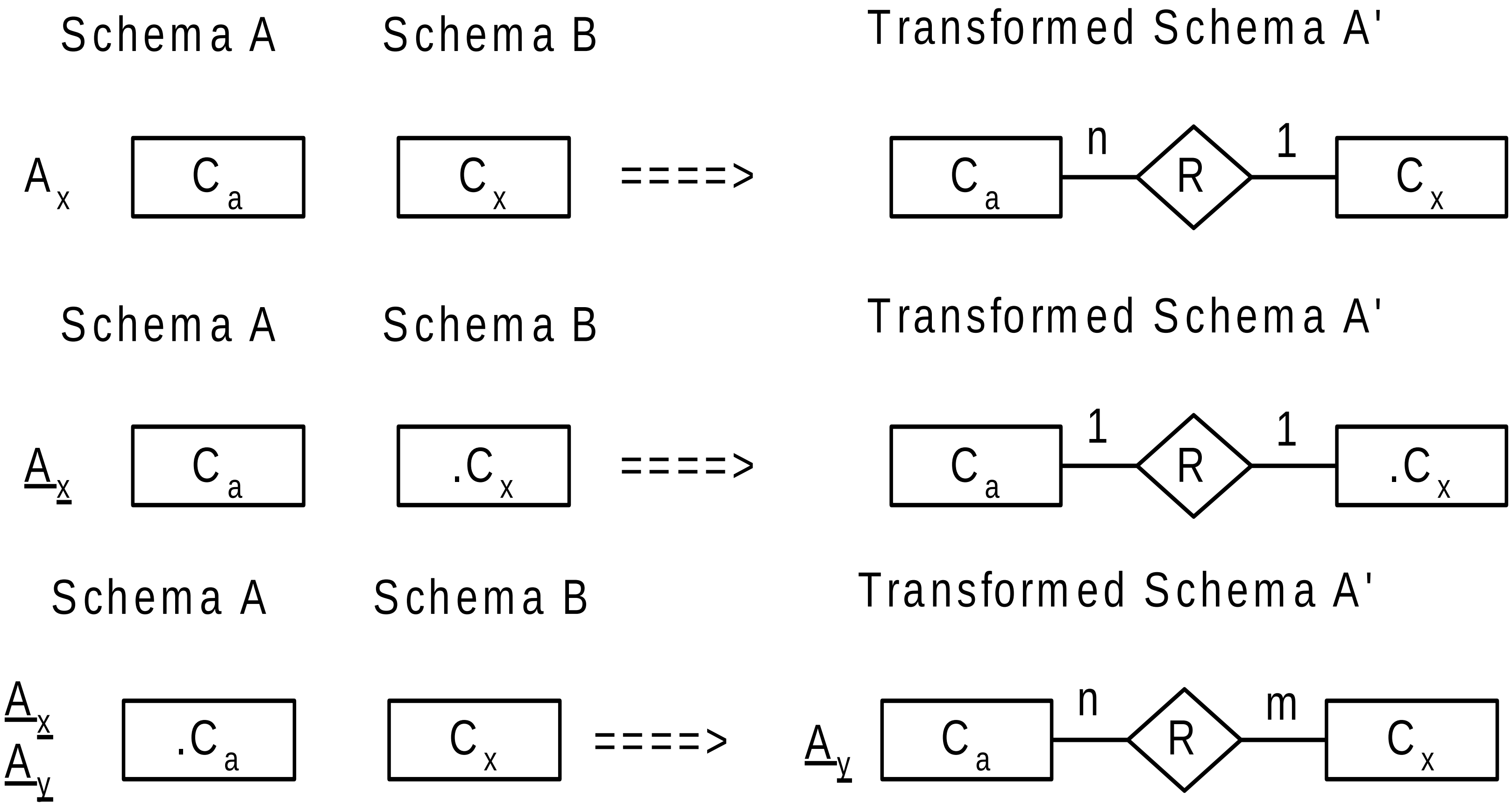


(note: Class$_a$ and Class$_b$ are synonyms, Attribute$_x$ are homonyms)

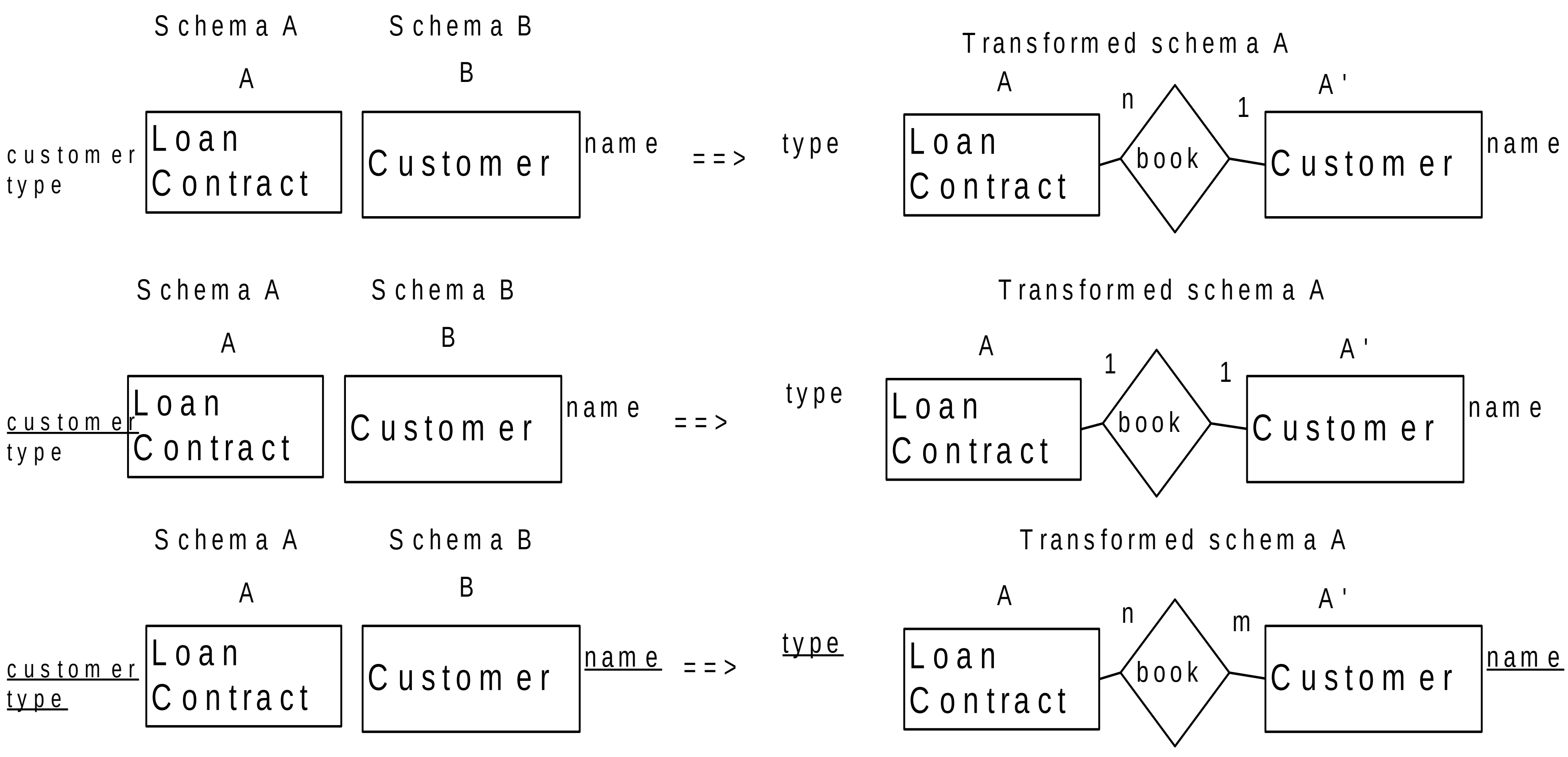
Example


Schema A
Schema B
Schema A'
Schema B'
Customer
Loan Borrower
====>
Customer
Customer
name
date
name
date
name
open-acct-date
name
loan-begin-date

Resolve conflicts on data types



Example


Schema A
Schema B
Transformed schema A
A
B
A
n
1
A'
customer type
Loan Contract
Customer
name
==>
type
Loan Contract
book
Customer
name
Schema A
Schema B
Transformed schema A
A
B
A
1
1
A'
customer type
Loan Contract
Customer
name
==>
type
Loan Contract
book
Customer
name
Schema A
Schema B
Transformed schema A
A
B
A
n
m
A'
customer type
Loan Contract
Customer
name
==>
type
Loan Contract
book
Customer
name

Resolve conflicts on key

Schema A

$\underline{A_y}$

A_z

C_x

Schema B

Cy_x

A_y

$\underline{A_z}$

====> User supervision

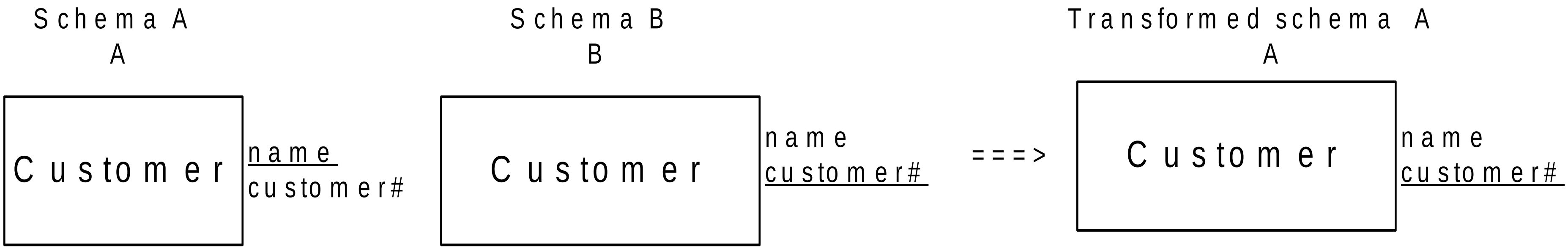
Transformed Schema X

A_y

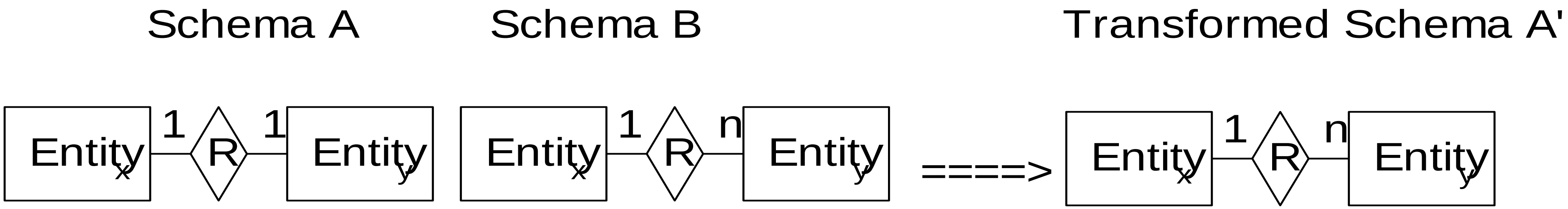
$\underline{.A_z}$

C_x

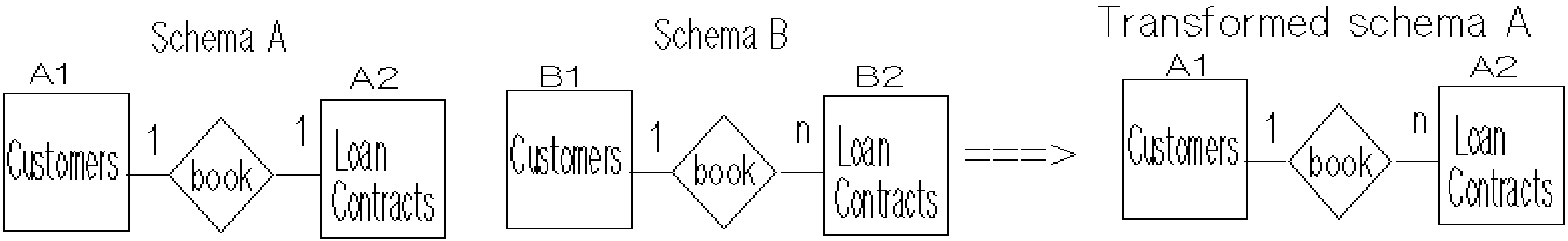
Example


Schema A
A
Customer
name
customer#
Schema B
B
Customer
name
customer#
===>
Transformed schema A
A
Customer
name
customer#

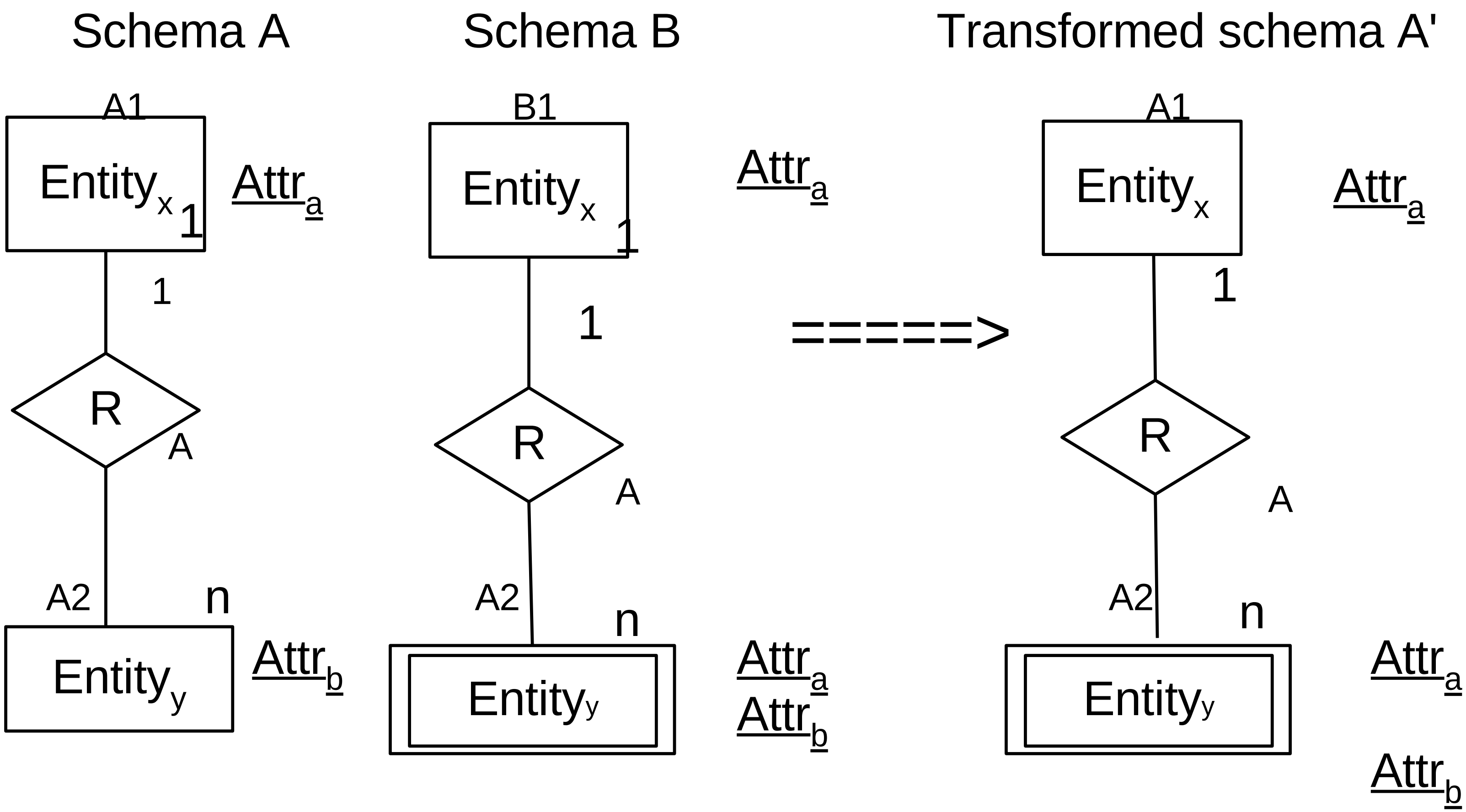
Resolve conflicts on cardinality



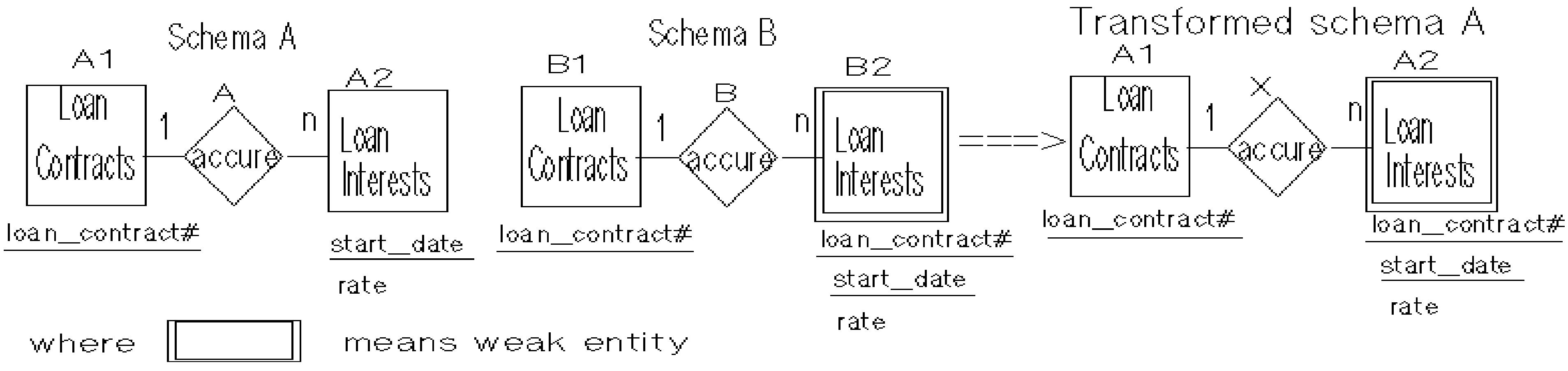
Example


Schema A
A1
Customers
1
book
1
A2
Loan Contracts
Schema B
B1
Customers
1
book
n
B2
Loan Contracts
===>
Transformed schema A
A1
Customers
1
book
n
A2
Loan Contracts

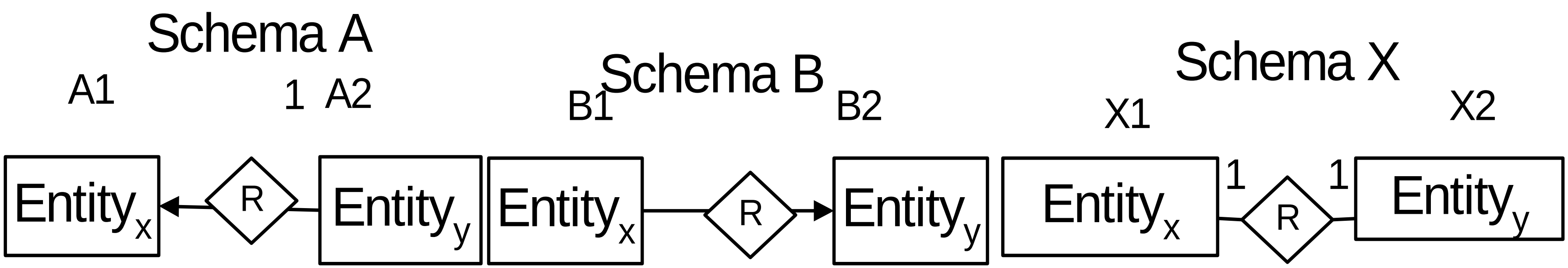
Resolve conflicts on weak entity



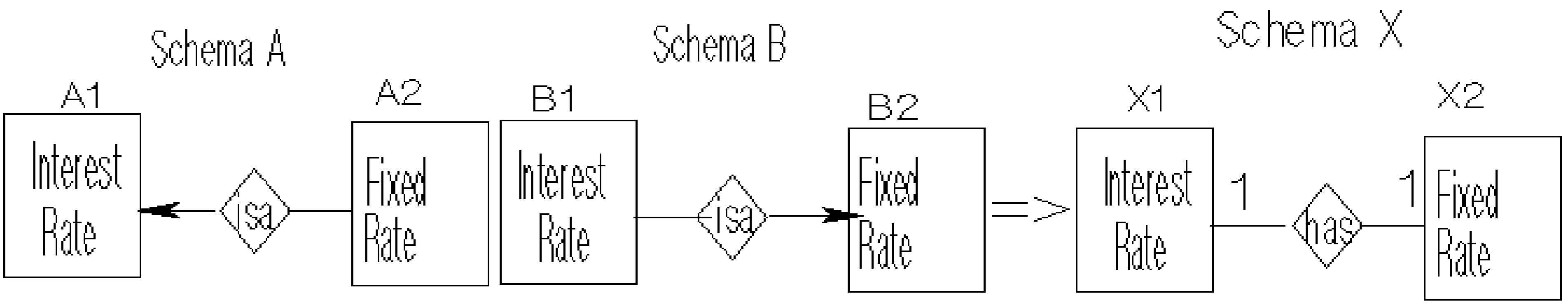
Example


Schema A
A1
Loan Contracts
1
A
accure
n
A2
Loan Interests
loan_contract#
start_date
rate
Schema B
B1
Loan Contracts
1
B
accure
n
B2
Loan Interests
loan_contract#
loan_contract#
start_date
rate
===>
Transformed schema A
A1
Loan Contracts
1
X
accure
n
A2
Loan Interests
loan_contract#
loan_contract#
start_date
rate
where
means weak entity

Resolve conflicts on subtype entity



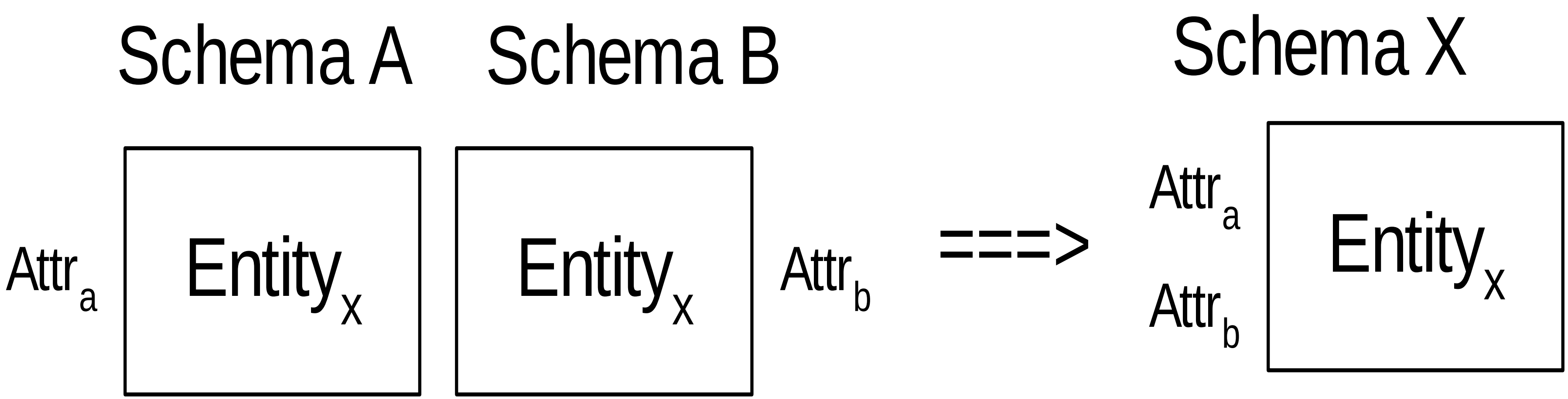
Example



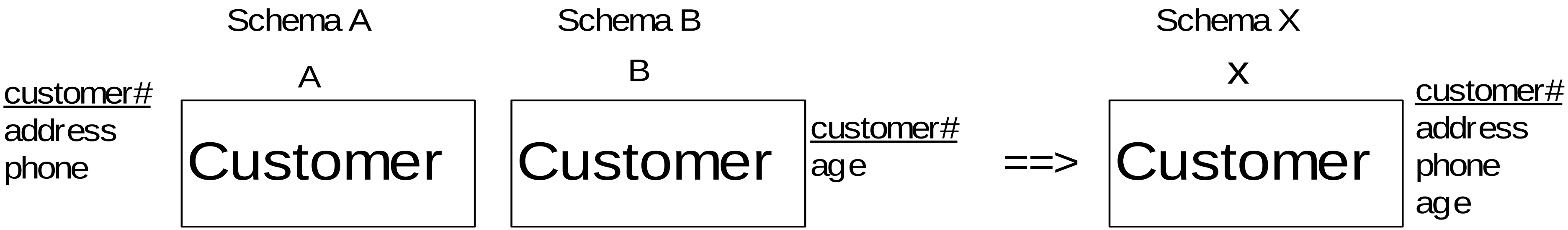
Schema A
Schema B
Schema X
A1
Interest Rate
isa
A2
Fixed Rate
B1
Interest Rate
isa
B2
Fixed Rate
=>
X1
Interest Rate
1
has
1
X2
Fixed Rate

Step 2　Merge entities

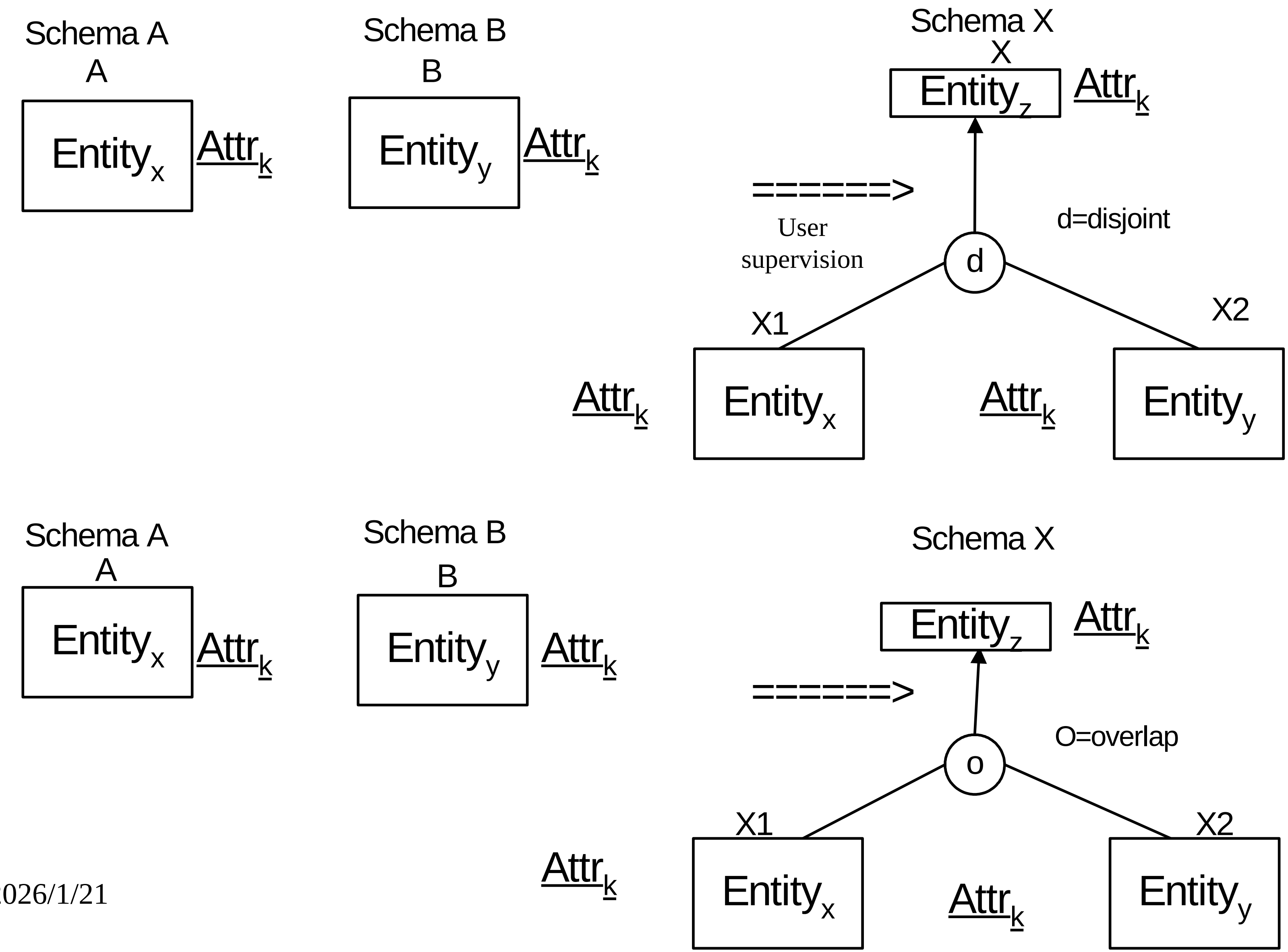
Merge classes by Union



Example


Schema A
Schema B
Schema X
A
B
X
customer#
address
phone
Customer
Customer
customer#
age
==>
Customer
customer#
address
phone
age

Merge EER models by Generalization



Example

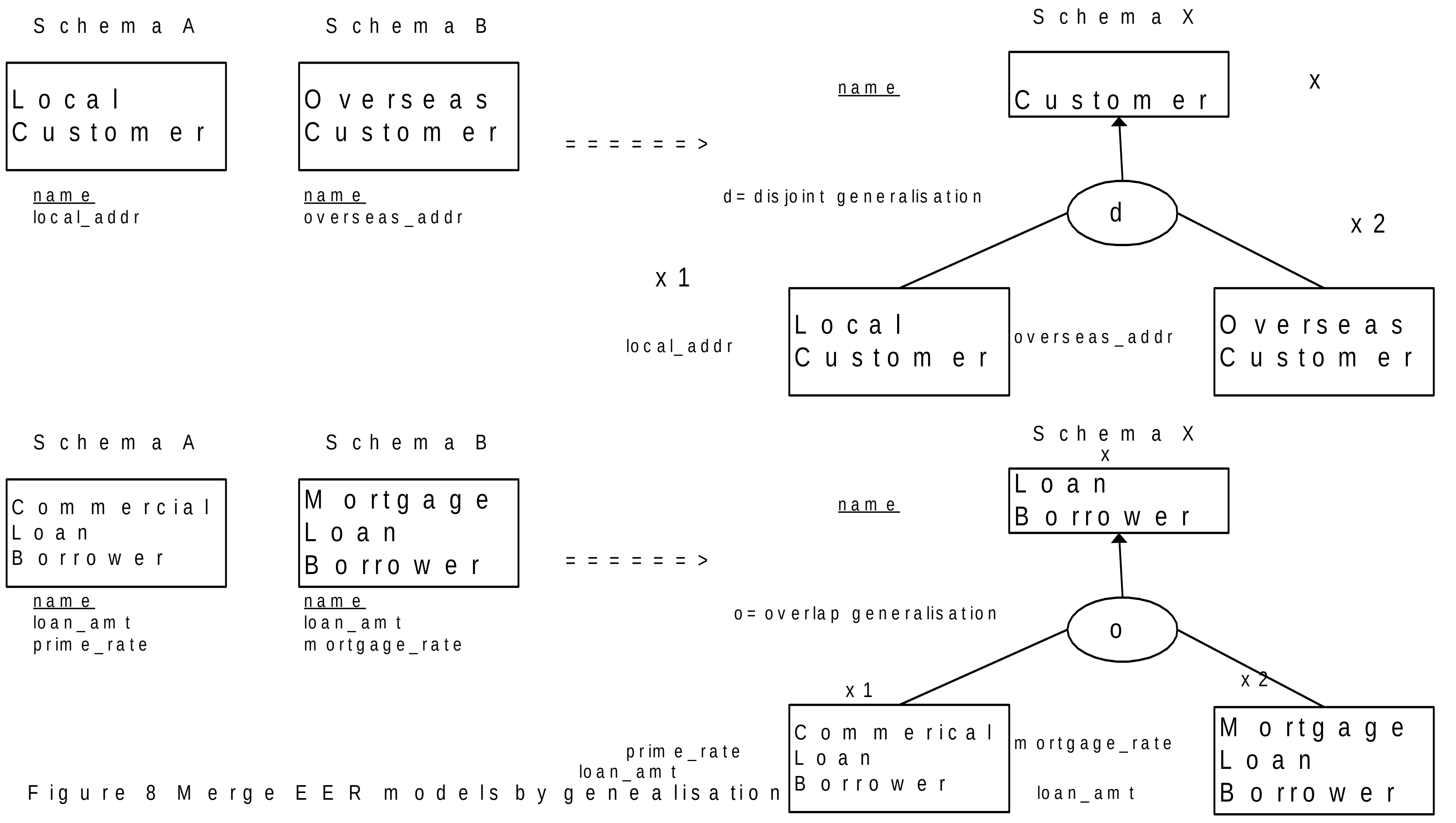
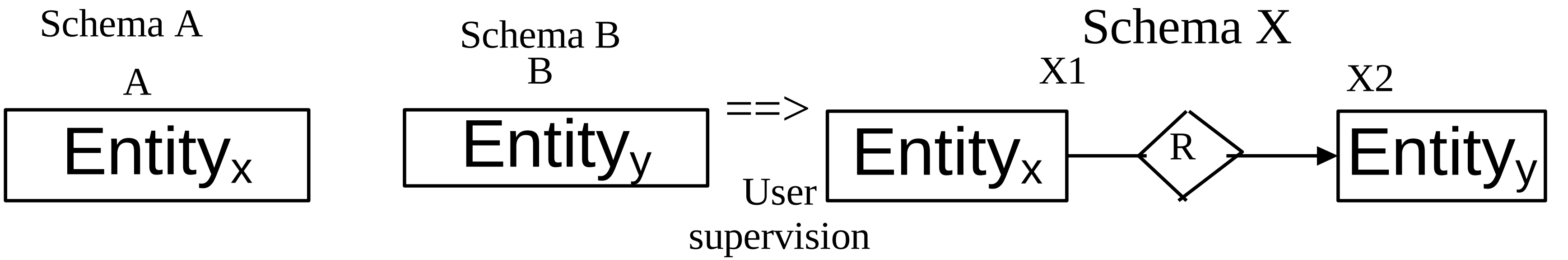
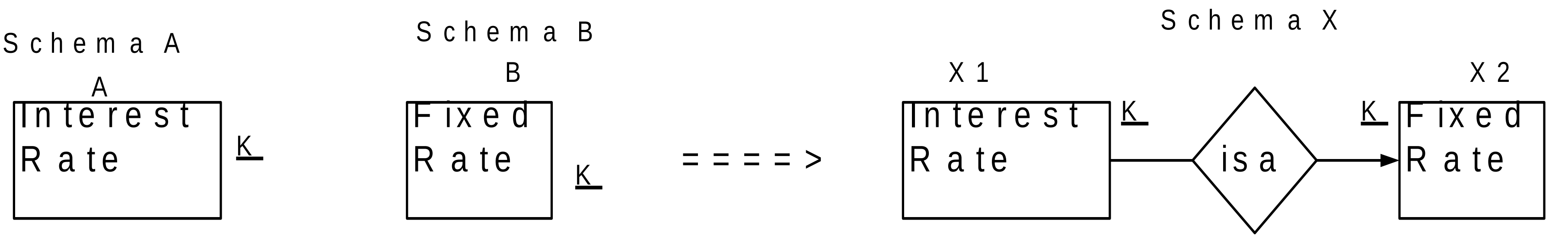


Figure 8 Merge EER models by genealisation

Merge EER models by Subtype Relationship

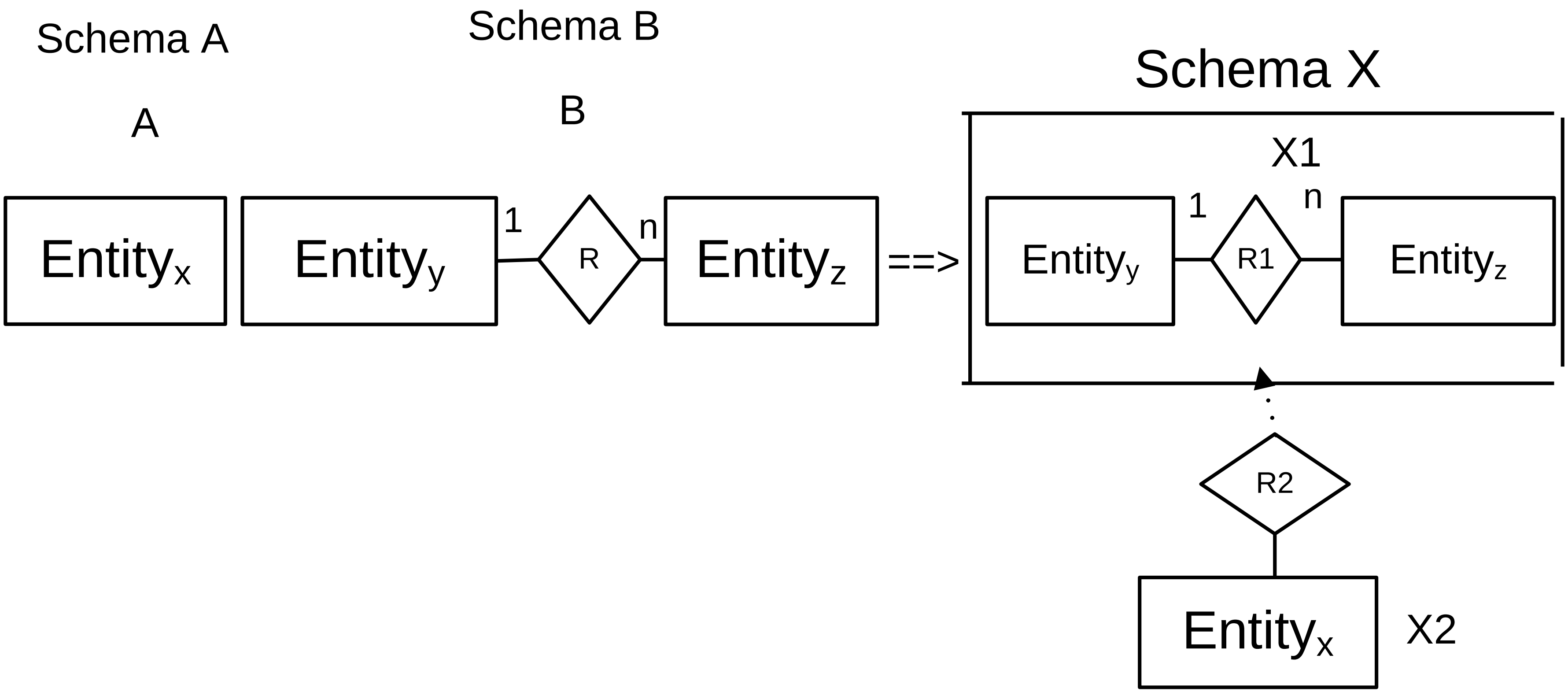


Example

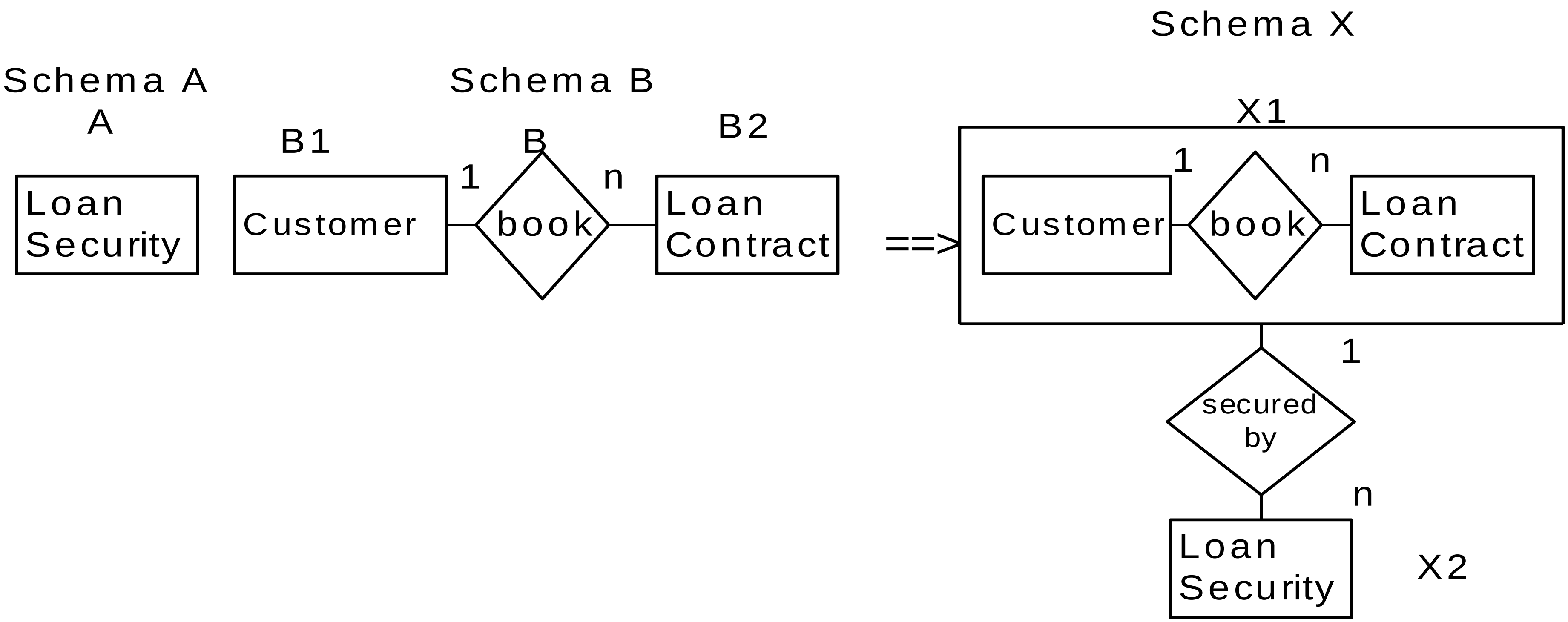


Schema A
A
Interest Rate
K
Schema B
B
Fixed Rate
K
=====>
Schema X
X1
Interest Rate
K
isa
K
X2
Fixed Rate

Merge entities by Aggregation

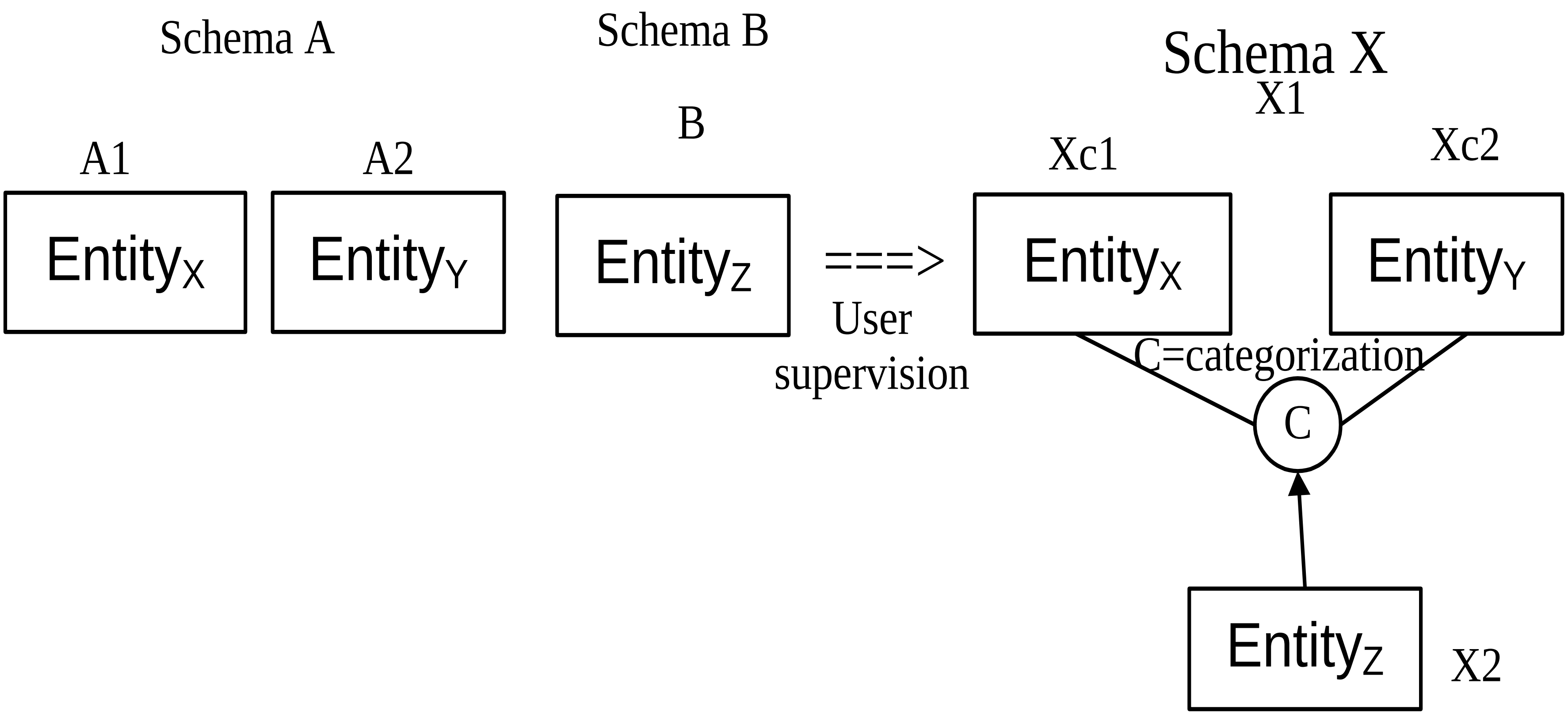


Example

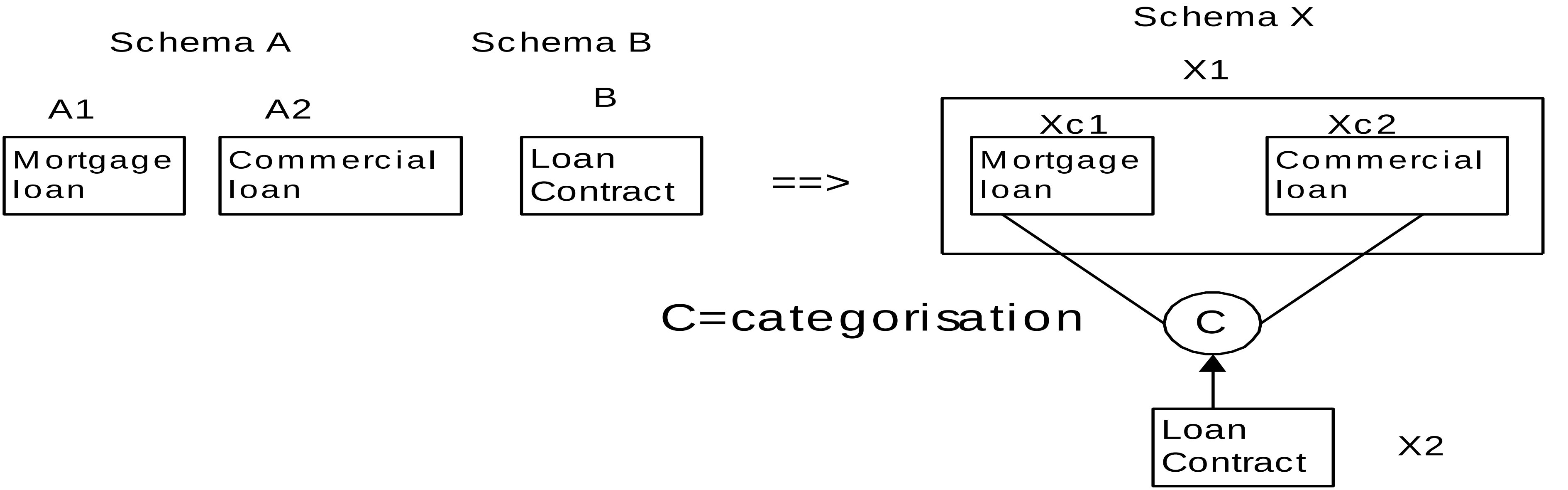


Schema X
Schema A
A
Schema B
B1
B
B2
X1
1
n
Loan
Security
Customer
book
Loan
Contract
==>
1
n
Customer
book
Loan
Contract
1
secured
by
n
Loan
Security
X2

Merge entities by categorization

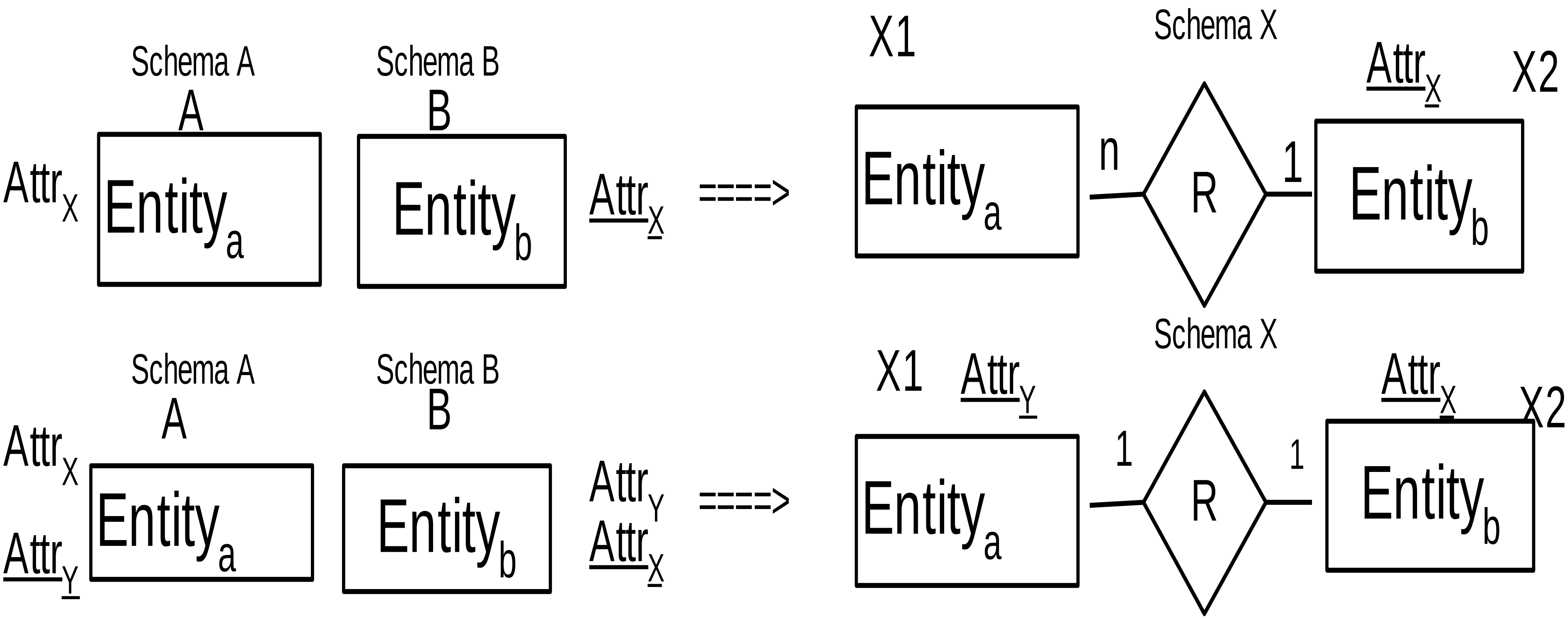


Example

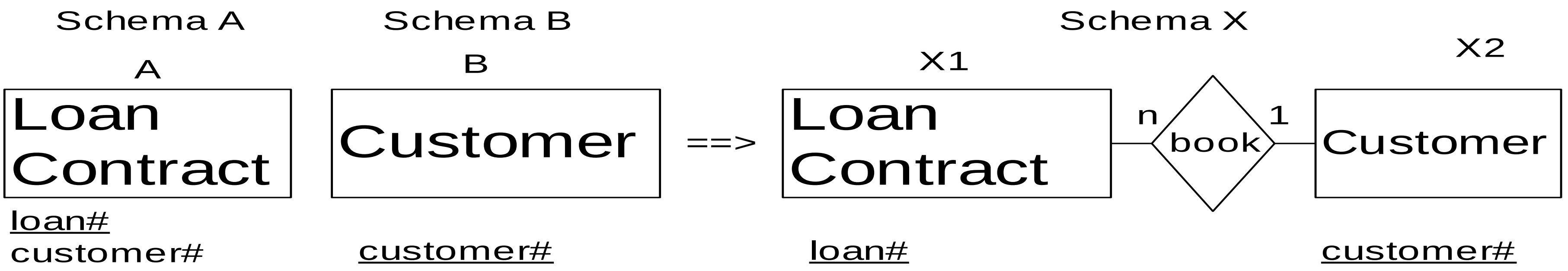


Schema A
Schema B
Schema X
A1
A2
B
X1
Mortgage loan
Commercial loan
Loan Contract
==>
Xc1
Xc2
Mortgage loan
Commercial loan
C=categorisation
C
Loan Contract
X2

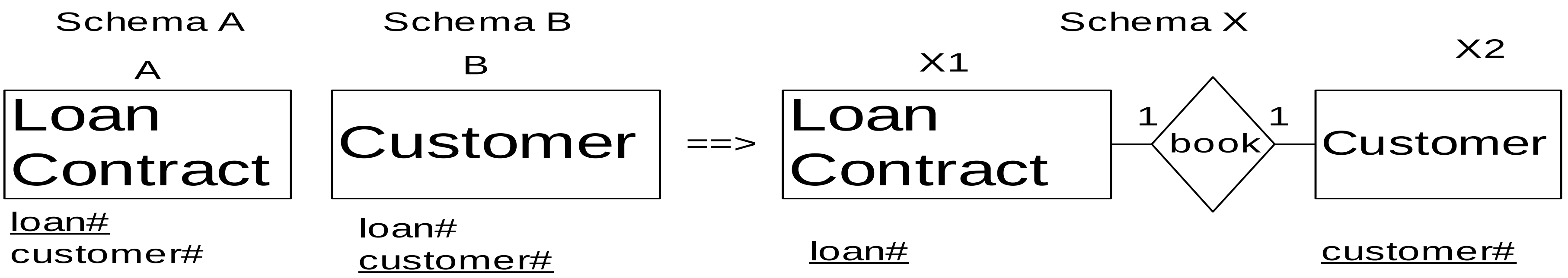
Merge entities by Implied Binary Relationship



Example



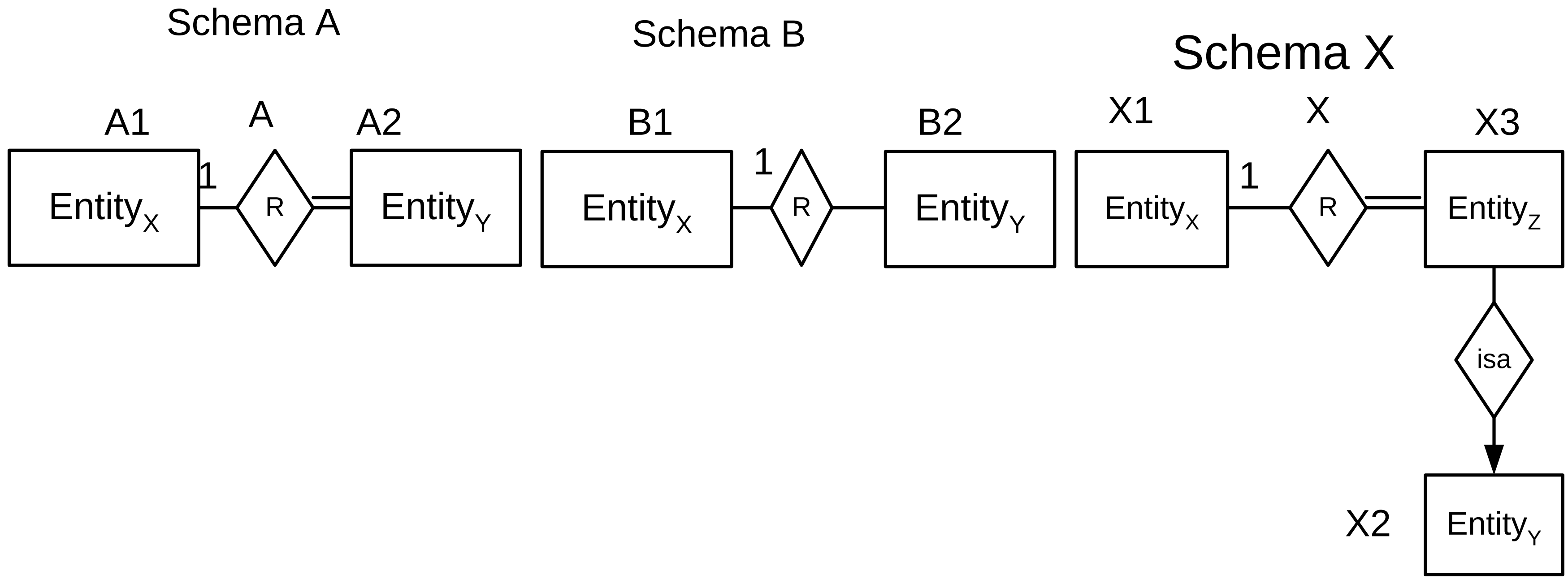
Schema A
Schema B
Schema X
A
B
X1
X2
Loan Contract
Customer
==>
Loan Contract
n
book
1
Customer
loan#
customer#
customer#
loan#
customer#



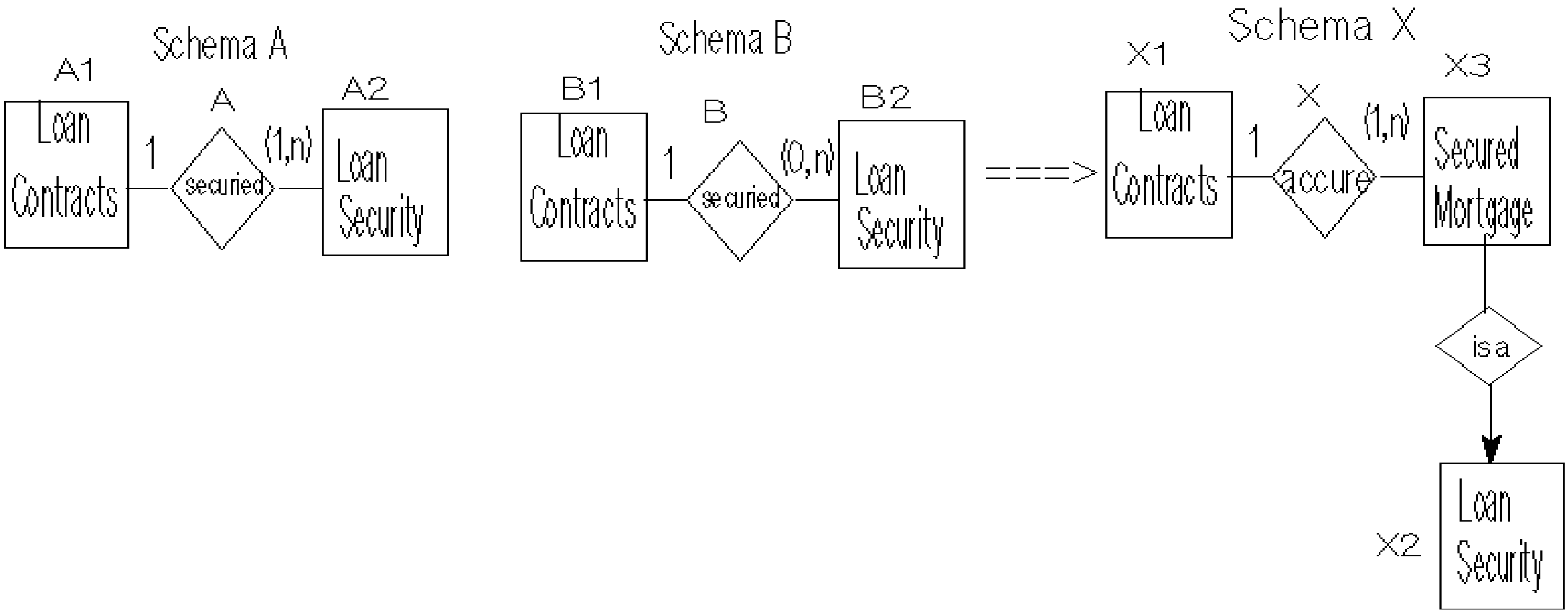
Schema A
Schema B
Schema X
A
B
X1
X2
Loan Contract
Customer
==>
Loan Contract
1
book
1
Customer
loan#
customer#
loan#
customer#
loan#
customer#

Step 3 Merge relationships

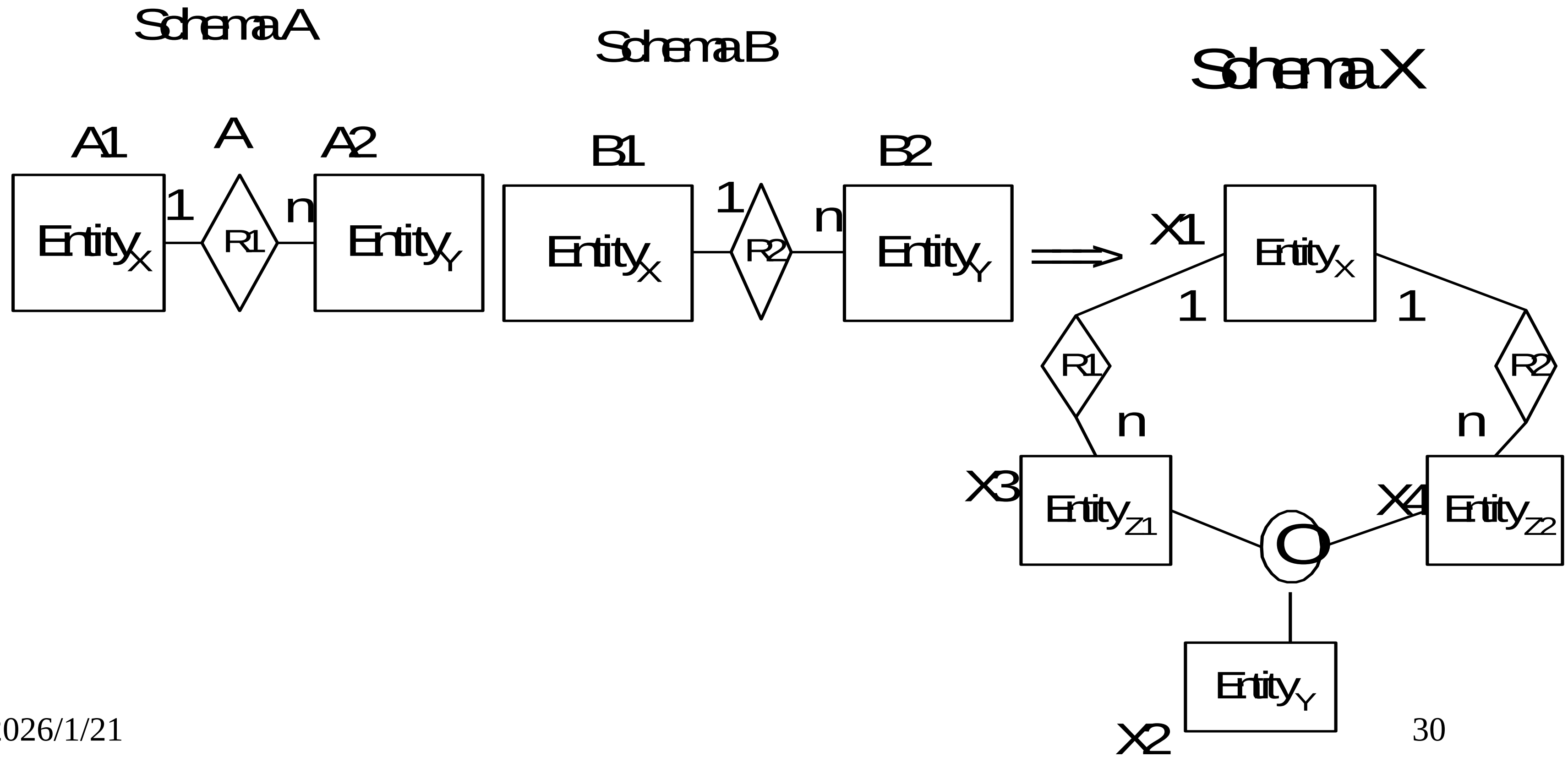
Merge relationships by subtype relationship



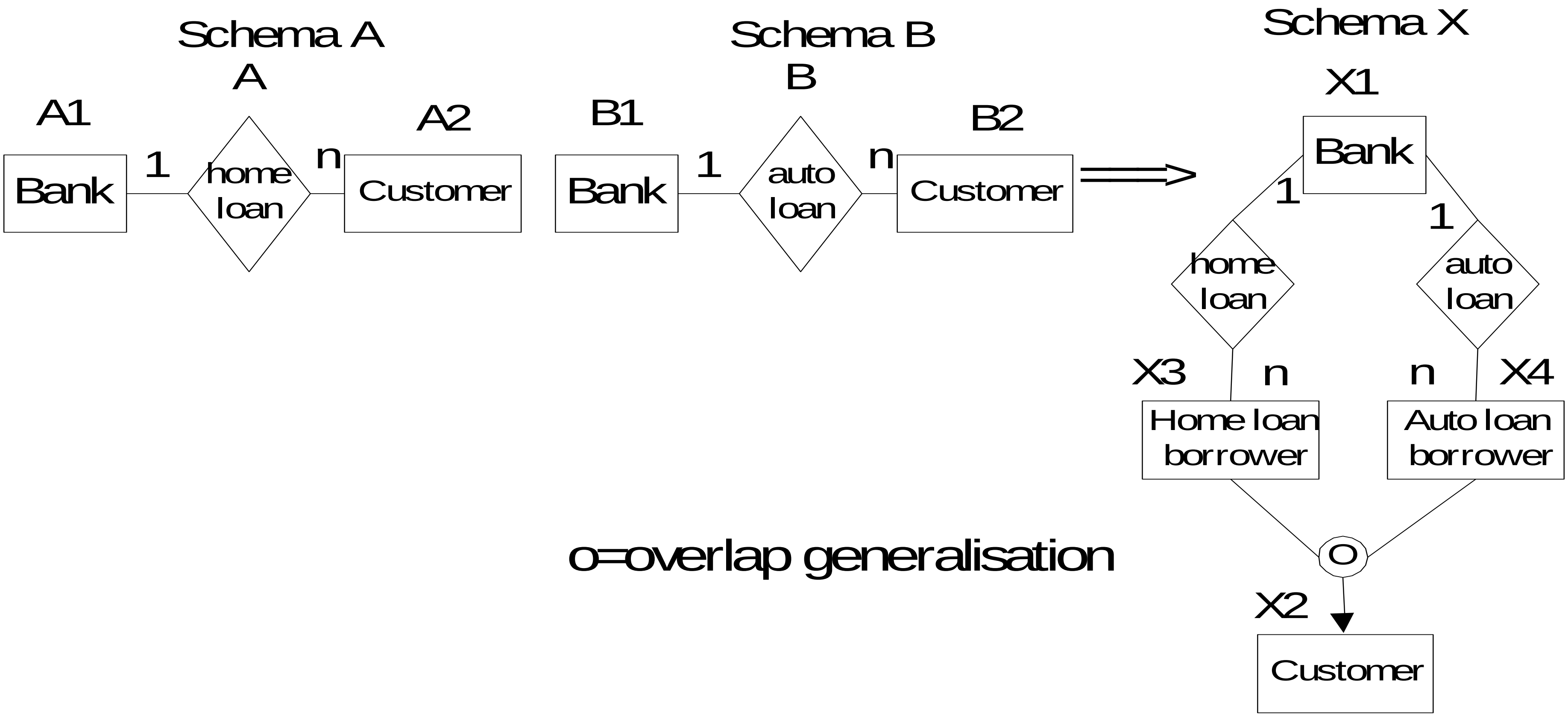
Example


Schema A
A1
Loan Contracts
1
A
securied
(1,n)
A2
Loan Security
Schema B
B1
Loan Contracts
1
B
securied
(0,n)
B2
Loan Security
===>
Schema X
X1
Loan Contracts
1
X
accure
(1,n)
X3
Secured Mortgage
is a
X2
Loan Security

Merge relationships by overlap generalization

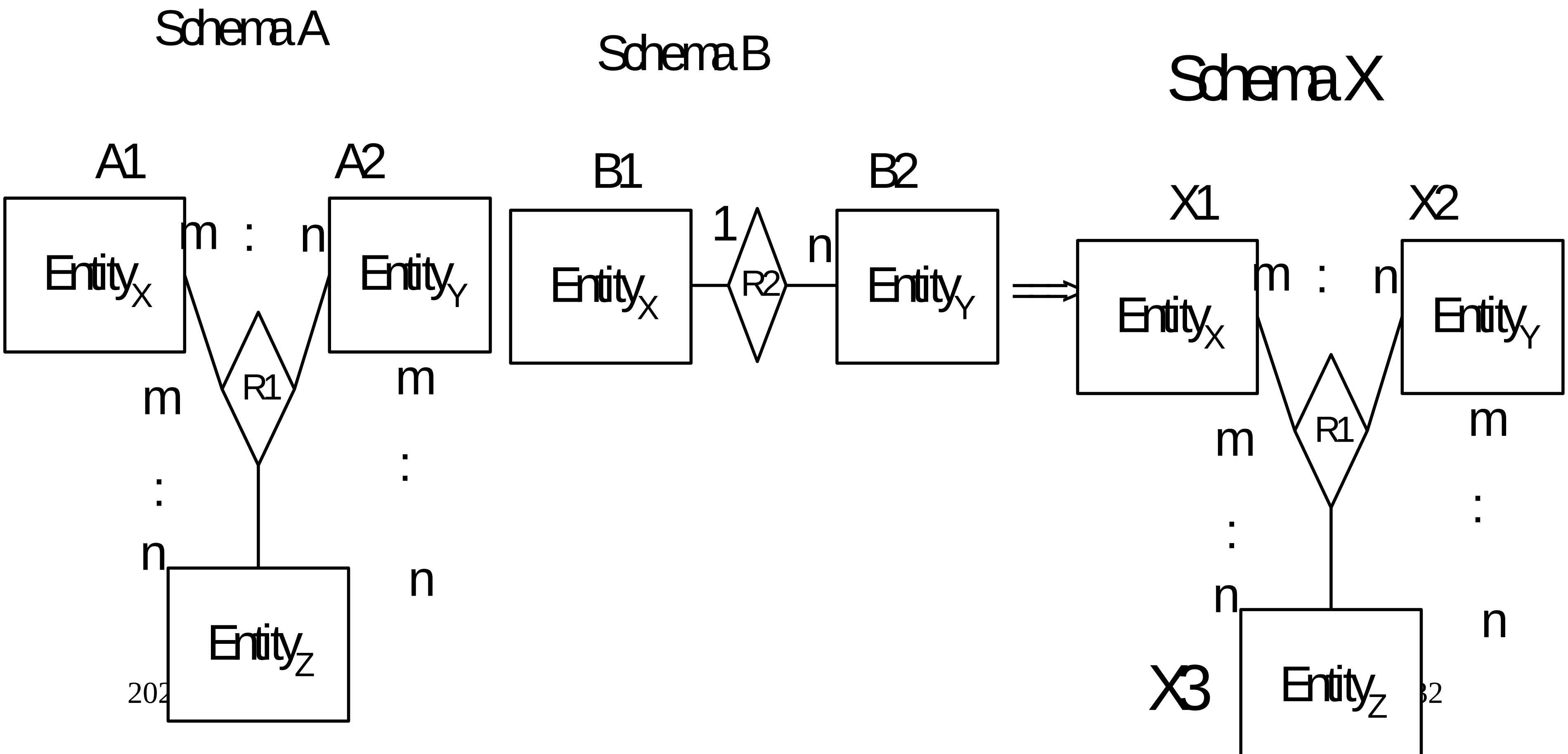


Example

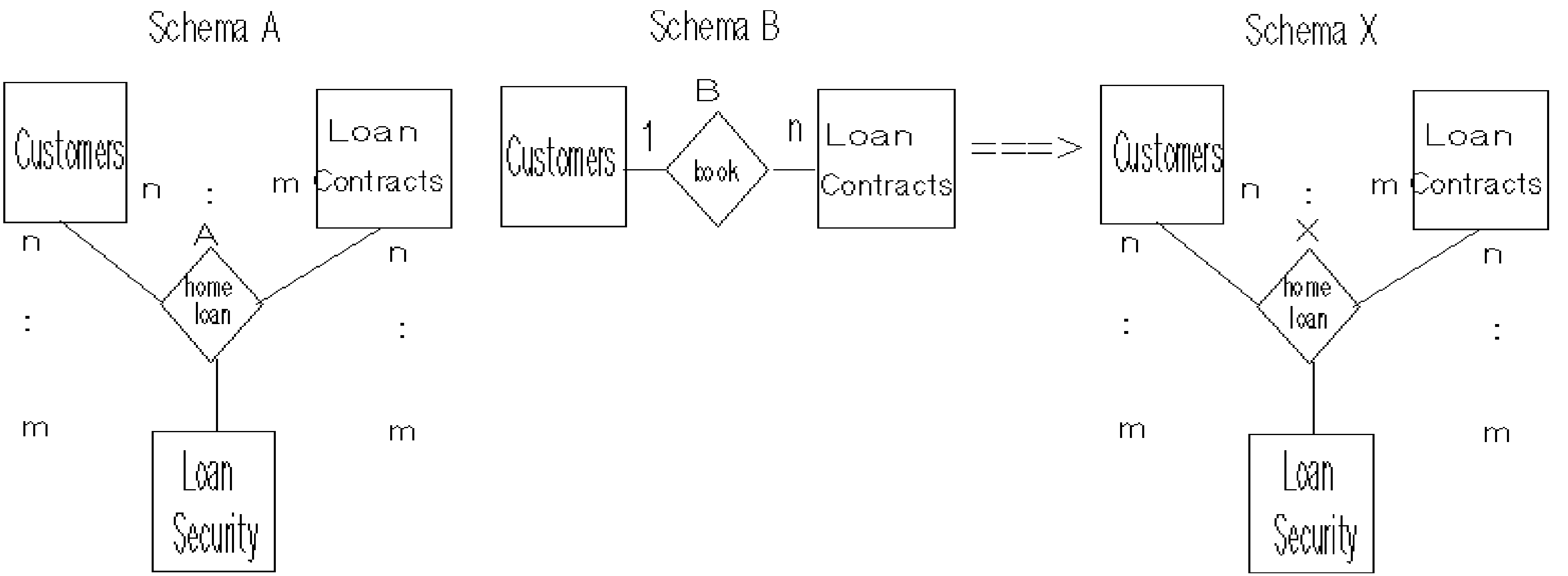


o=overlap generalisation

Absorbing Lower degree Relationship into a Higher degree Relationship

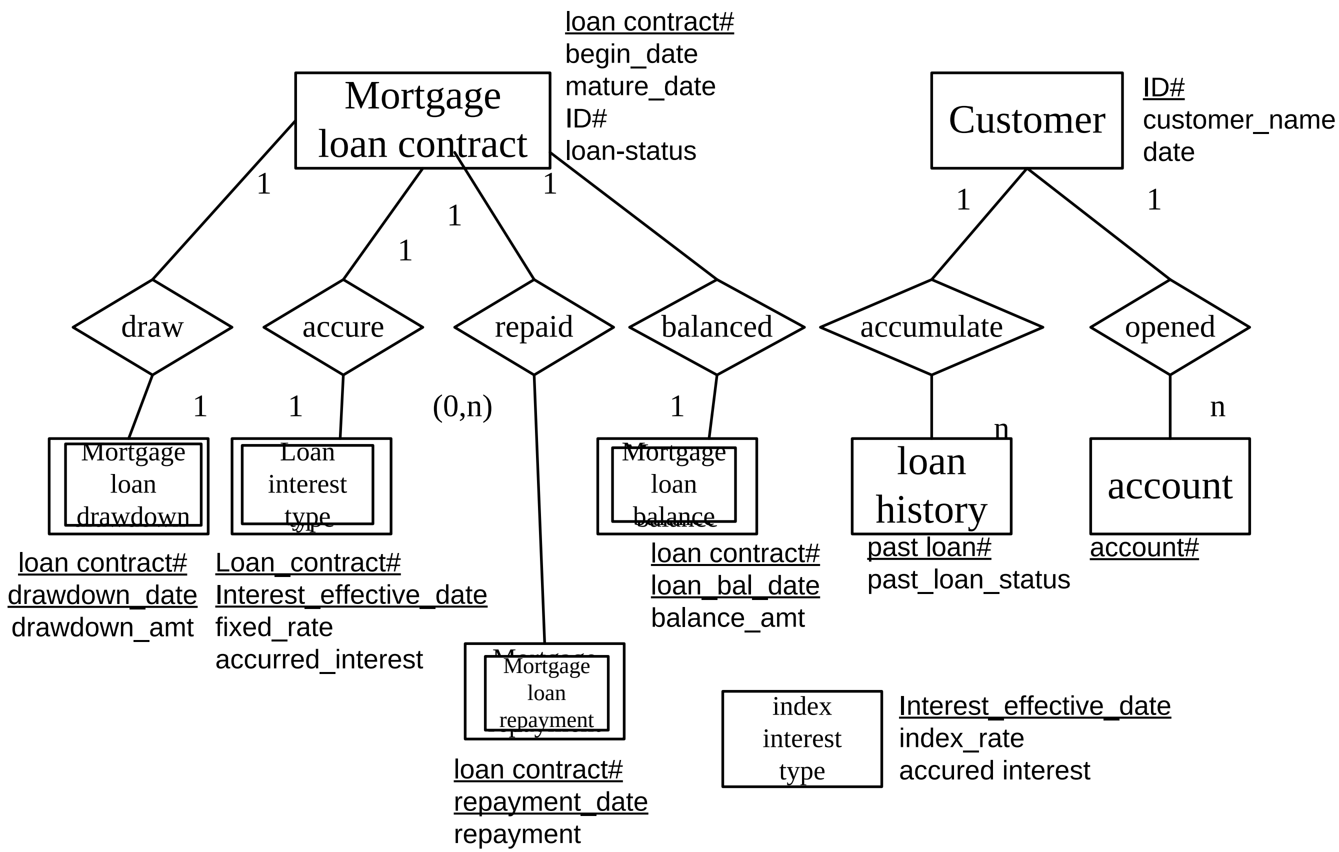


Example



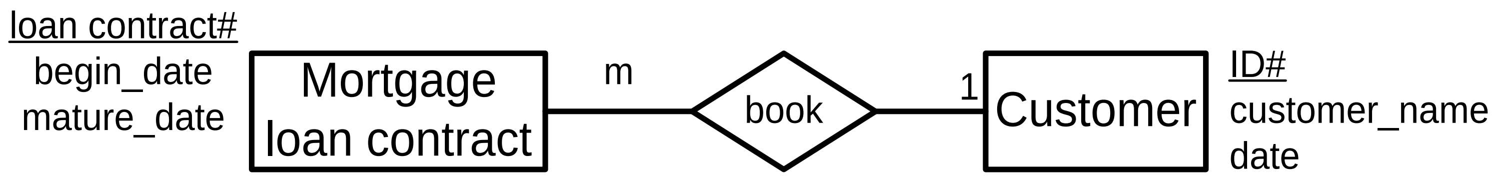
Schema A
Customers
Loan Contracts
home loan
A
n
:
m
n
:
m
n
:
m
Loan Security
Schema B
Customers
B
book
1
n
Loan Contracts
===>
Schema X
Customers
Loan Contracts
home loan
X
n
:
m
n
:
m
n
:
m
Loan Security

Case Study: In a bank, there are existing databases with different schemas: one for the customers, another for the mortgage loan contracts and a third for the index interest rate. However, there is a need to integrate them together for an international banking loan system.

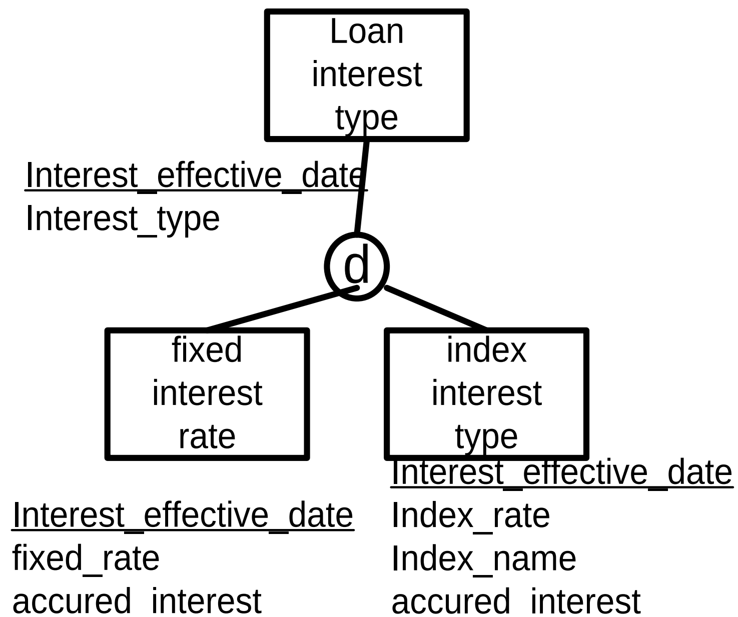


Step 2 Merge entities by Implied binary relationship and Generalization

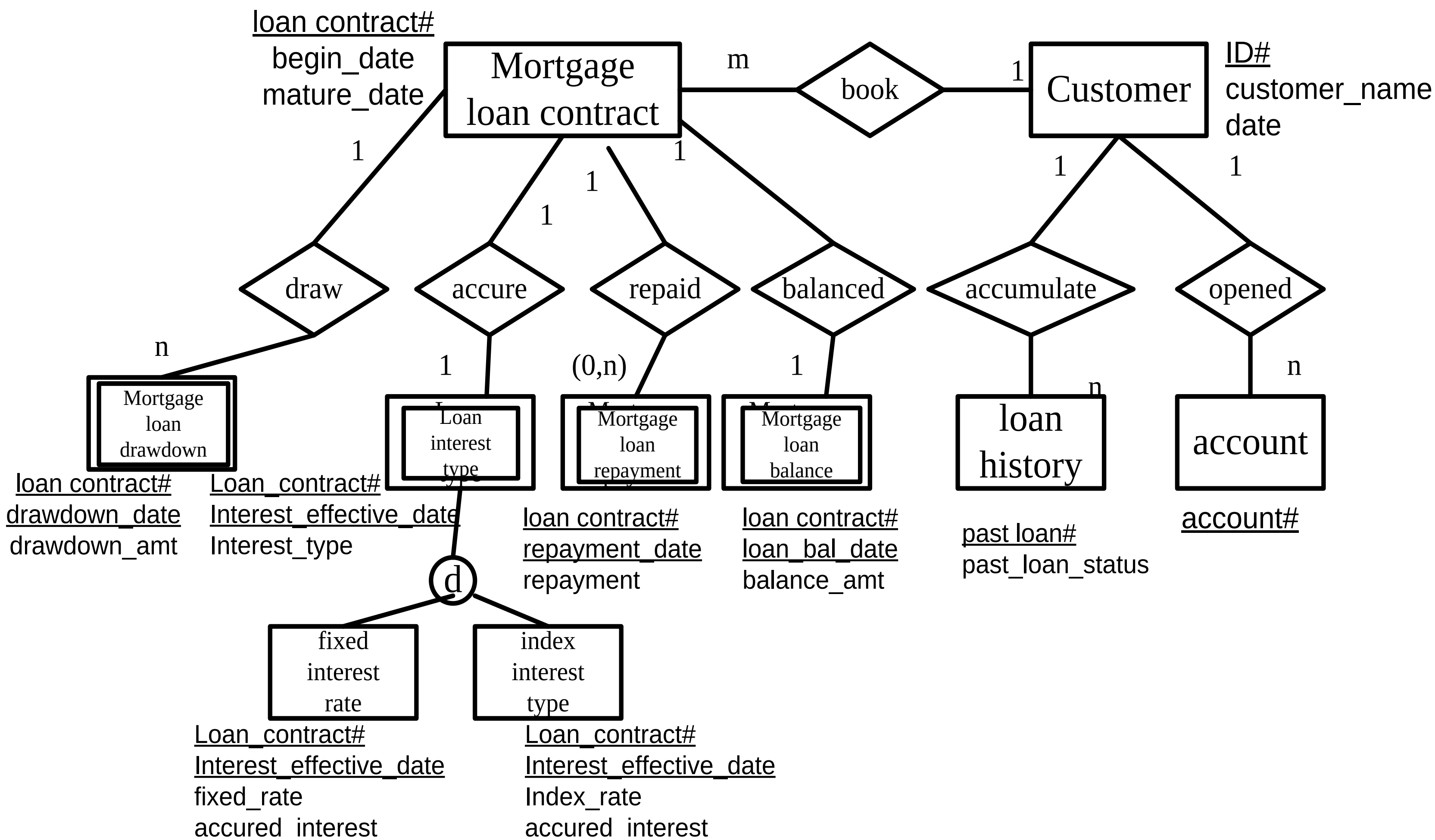
Since data ID# appears in entity Mortgage Loan Contract and entity Customer, they are in one-to-many cardinality with foreign key on the "many" side.



Since Fixed and Index Interest Rate both have the same key, Interest_effective_date, they can be generalized disjointed as either fixed or index rate.



Thus, we can derive cardinality from the implied relationship between these entities, and integrate the two schemas into one EER model.



Reading assignment

"Information Systems Reengineering and Integration" by Joseph Fong, published by Springer Verlag, 2006, pp. 282-310.

Review question 3

Can two Relational Schemas be integrated into one Relational Schema? Explain your answer.

Which steps need user supervision for integrating two Extended Entity Relationship Models into one Extended Entity Relationship Model? Explain your answer.

Tutorial question 3

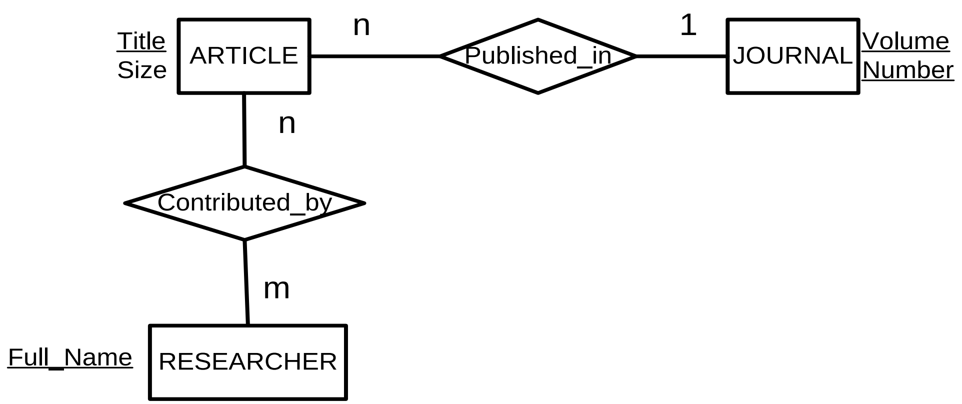
Provide an integrated schema for the following two views which are merged to create a bibliographic database. During identification of correspondences between the two views, the users discover the followings:

RESEARCHER and AUTHOR are synonyms,
CONTRIBUTED_BY and WRITTEN_IN are synonyms,
ARTICLES belongs to a SUBJECT.
ARTICLES and BOOK can be generalized as PUBLICATION.

Hints: Given two subclass entities have same relationship(s). The two subclasses entities can be generalized into a

superclass entity and the subclass relationship(s) can also be generalized into a superclass relationship..

View 1



View 2